



Explore America

Interactive Multimedia Web Application

CS 430 – Internet Multimedia Programming
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waril2001.github.io/Explore-america-multimedia

Project Overview

What is Explore America?

A travel guide web app helping users discover two iconic American destinations — Seattle, WA and San Juan, PR — based on their personal travel preferences.

Users select a climate preference and get routed to a destination page with rich multimedia content: videos, image galleries, ambient audio, and a personalized day-by-day trip planner.



Live Site

GitHub Pages deployment



Responsive

Mobile-friendly CSS layout



Accessible

ARIA, skip links, focus styles

[Home](#)[Seattle](#)[San Juan](#)[Planner](#)[About](#)[Credits](#)

Key Features — Destination Pages

Home Page — Destination Selector

Two climate buttons (Cold & Urban / Warm & Tropical) route users to matching city pages. A recommendation preview box appears before redirecting.

Seattle Page

Hero image + overlay card, video tour with controls & fallback, image gallery (Space Needle, Pike Place, Waterfront) with hover zoom, fun facts via JavaScript DOM, and an ambient audio player with ARIA labels.

San Juan Page

Beach background, about section, gallery of 3 attractions (El Morro, Old San Juan, Condado Beach), fun facts button, and a CTA directing users to the trip planner.

Seattle Page — HTML Gallery Code

Semantic Gallery Structure

```
<section class="gallery">
```

Semantic grouping — screen readers identify this as a distinct region.

```
alt text on every <img>
```

Descriptive alt strings describe the image for screen readers, not just the filename.

```
<p> caption below each image
```

Text label visible to all users, paired with the image for context.

```
object-fit: cover in CSS
```

Keeps all gallery images the same 250×150px crop regardless of original dimensions.

```
transform: scale() on :hover
```

Subtle zoom effect gives visual feedback without JavaScript — pure CSS.

seattle.html — Gallery Section

```
<section class="gallery">
  <h3>Top Attractions</h3>
  <div class="gallery-grid">
    <div>
      
      <p>Space Needle</p>
    </div>
    <!-- Pike Place, Waterfront
      follow same pattern -->
  </div>
</section>
```

Key Feature — Trip Planner

The Most Interactive Feature

- Name, destination, days (1–7), budget & interests
- Client-side validation with inline error messages
- Interest-based itinerary: 2 activities/day from curated list
- Budget tip displayed at itinerary bottom
- Reset / 'Plan Another Trip' button

planner.html — Form Snippet

```
<select id="destination"
  name="destination"
  required
  aria-required="true">
  <option value="">-- Select city --</option>
  <option value="seattle">
    Seattle, WA
  </option>
</select>
<span class="error-msg"
  id="destError"
  role="alert">
  Please select a destination.
</span>
```

JavaScript Interactivity

Destination Selector

Event listener detects click, shows recommendation box, then redirects after 2s timeout.

Form Validation

Checks each field on submit. Injects error messages via DOM; uses role='alert' for screen readers.

Itinerary Generator

Loops through selected interests, maps them to activity pools, builds day-by-day HTML dynamically.

main.js — Dynamic Fun Facts Toggle

```
const btn = document.querySelector('#funFactsBtn');
btn.addEventListener('click', () => {
  const box = document.querySelector('#factsBox');
  box.classList.toggle('hidden'); // Show/hide dynamically
  box.setAttribute('aria-live', 'polite'); // Announces to screen readers
});
```

CSS — Design Decisions & Structure

Color System

#1e3a5f / #fca311

Navy is the primary brand color on all headers, nav, buttons, and cards. Orange is reserved for accents — hover states, active links, CTA borders — so it always draws the eye.

Hover & Transitions

`transition: transform 0.3s ease;`

Gallery images scale up 5% on hover. Nav links and buttons swap to orange. All transitions are CSS-only — no JS needed for visual feedback.

Accessibility — Focus

`outline: 3px solid #fca311;`

Every focusable element — links, buttons, inputs, selects, textareas — gets a visible orange outline on focus. This is required for keyboard navigation compliance.

Flexbox Layout

`display: flex; flex-wrap: wrap;`

Used on nav ul, gallery-grid, and checkbox groups. flex-wrap lets them reflow naturally on narrower screens without a separate mobile rule.

Responsive Breakpoints

`@media (max-width: 768px)`

Two breakpoints: 768px stacks the nav vertically, collapses the gallery to a column, and hides the floating GIF. 480px shrinks fonts and stretches buttons to full width.

Component Styles

`.planner-form-wrap, .team-card`

Self-contained blocks with white backgrounds, border-radius: 14px, and box-shadow. Each page's unique components (planner, credits table, about cards) are scoped in their own CSS section.

Technical Decisions — HTML Structure

San Juan — Video Section

```
<section class="video-section">
  <video autoplay muted loop
    controls playsinline
    aria-label="Video tour
      of San Juan, PR">
    <source src="media/video/PR.mp4"
      type="video/mp4">
    <a href="media/video/PR.mp4">
      Download video</a>
  </video>
</section>
```

- autoplay muted loop — plays silently on load without annoying the user
- playsinline — prevents iOS from hijacking to fullscreen
- aria-label — describes the video for screen readers
- <source> inside <video> — allows multiple formats for browser compatibility
- Fallback <a> link — users on unsupported browsers can still access the video

Semantic HTML5 Choices

<header>, <nav>, <main>

Landmark roles — screen readers jump between them

<section>, <article>

Meaningful content regions vs decorative divs

<footer>

Consistent site-wide footer with nav links

aria-label on media

Every <video> and <audio> is described

role="alert" on errors

Form errors announced immediately by screen readers

Skip link (.skip-link)

Lets keyboard users jump past repeated nav directly to main

novalidate on <form>

Suppresses browser-native popups; we control all UI/messages

GitHub Collaboration Workflow

1

Repo Setup

Public repo initialized with README & .gitignore

2

Feature Branches

Each member on their own branch (e.g. feature/about-contact-pages)

3

Regular Commits

Descriptive commit messages after each meaningful change

4

Pull Requests

PRs opened; teammates added as reviewers before merging to main

5

GitHub Pages

Auto-deploys from main — live within minutes of merge



waril2001.github.io/Explore-america-multimedia | Changes to main auto-deploy via GitHub Pages

Challenges & Solutions



Meeting All Requirements

→ Reviewed project guidelines multiple times; kept a checklist for semantic HTML, JS interactivity, multimedia, accessibility, and ethical media use.



Trip Planner JavaScript Logic

→ Multi-field validation, checkbox groups, interest-to-activity mapping, and dynamic day cards required significant iteration and debugging.



File Structure & Asset Management

→ Reorganized two separate asset folders into one media/ folder with subfolders (images/, video/, audio/, animations/) and updated all path references.



Accessibility Implementation

→ Beyond alt text: added skip links, ARIA attributes, keyboard focus styles, and aria-live regions for screen-reader-compatible dynamic content.



GitHub Collaboration

→ Adopted feature branches + PR workflow. Careful coordination prevented merge conflicts and overwrites across team members.



Live Demonstration

waril2001.github.io/Explore-america-multimedia

- Home destination selector flow
- Seattle page — video, gallery, audio player
- San Juan page — hero, gallery, fun facts
- Trip Planner — form validation + itinerary output
- Contact form — validation & success message

Class Engagement — Which Destination Are You?



Quick Poll — Vote with your vibe!



SEATTLE

Cold & Urban

- ☁️ You love moody, rainy days
- ☕ Coffee is your love language
- 🎸 Indie music & tech culture
- ⛰️ Mountains & ocean views

Raise your left hand! 🖐️



SAN JUAN

Warm & Tropical

- ☀️ Beach & sunshine all year
- 💃 Salsa, reggaeton & good food
- 🏰 History + colorful streets
- 🌺 Island vibes only

Raise your right hand! 🖐️

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